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Step Change in West of Shetland Offshore Fracturing. First Online Multistage Fracs in Clair Delivers Operational Efficiency and Production Uplift

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Abstract

In the Greater Clair Area, the largest oil field in Europe, Hydraulic Fracturing (HF) has demonstrated its value in terms of delivering initial production uplift increases of 3 to 6 times with the first HF wells being installed in Clair Phase 1, 2019, and Clair Ridge, 2022. Clair Ridge platform started production in 2018; however, some wells have not encountered sufficient natural fractures (NF) to replicate oil production rates from the initial appraisal wells, increasing the appetite for HF.

SPE 216632 (Bird, 2023) documented successful production delivery from intervention-based HF in Clair, while highlighting these operations were inefficient on platforms not specifically designed to support HF operations. SPE 216632 expressed the case to move away from delivering HF in Clair as intense and complex well work campaigns to instead efficiently deliver HF ""online"" via drill pipe as part of the lower completion installation.

This paper will document the successful planning, design, and execution of the first ""online"" HF completion operation delivered for Clair Ridge. 5 HF were executed, including record breaking efficiency for Clair with 4 treatments in 4 days. This entire process will be reviewed from completion concept selection, supplier / equipment selection, completion design, operational execution, and lessons learned.

The paper will document the potential to incorporate sleeves designed for hydraulic fracturing in addition to multiple additional sleeves designed to provide proppant retention, sand control, and remote activated sleeves for intervention-less production start-up. Providing such functionality currently requires two additional sleeves per zone (e.g., increasing from 8 to 24 sleeves in an 8 zone well). Work is underway to incorporate either 3 sleeves per zone or full functionality into a single sleeve. The pros and cons of each approach will be presented outlining the risks associated, such as accidental sleeve shifting or proppant blockage around sand control sleeve annulus.

Clair Unit III has historically been completed with passive sand control standalone screens (SAS) which was a borderline decision. This is the first Unit III well to be completed without SAS, with HF providing additional mitigation for this decision. The process to install a robust sand-face completion will

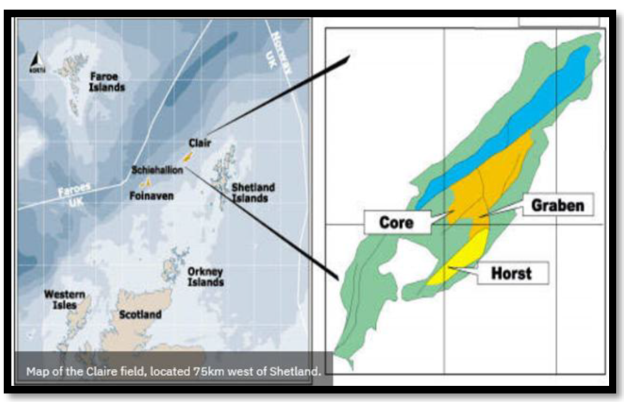
be documented. Details will be provided of solids production monitoring on startup through a temporary well clean-up package designed to remove solids and water and ensure smooth plant operations, including the use of Cutting Re-Injection (CRI) well to dispose of flow back fluids.

"Online" HF operations have been operationally and technically successful, the production performance predicted post-drill and actual production post HF will be reviewed including installing a remote activated downhole choke inserted to control production from a NF zone at risk of water production. Analysis of tracers to compare production splits between naturally fractured zones and HF zones will be presented.

Introduction

The Clair Oilfield and its reservoir characteristics have been the subject of extensive research and documentation across numerous publications (Ridd, 1981) (Allen, 1992), (Barr, 2007), (Coney, 1993), (Nichols, 2005), (Webster, 2018), (Roy, 2022), (Dumitrache, 2022), (Bird, 2023) For context, a concise overview is provided here. The field was first identified in 1977 through exploration well 206/08-1A, which encountered a substantial oil column within a thick sequence of Devonian to Carboniferous continental sandstones. Spanning over 200 square kilometers and containing an estimated 7 billion barrels of stock tank oil initially in place (STOIIP), Clair stands as the largest oilfield on the UK Continental Shelf. Despite its scale, the field presents considerable reservoir complexity, largely due to intricate natural fracture systems.

A lengthy appraisal phase including a 1996 extended well test demonstrated sustainable production rates of 15,000 to 20,000 barrels of oil per day from long horizontal wells. These wells targeted highly productive, naturally fractured zones aligned with mapped fault structures, proving instrumental in enabling the field's development. Development approval was received in 2001, with production commencing in 2005 via the Clair Phase 1 platform primarily targeting Horst and Graben regions. Clair Ridge platform (online 2018), focuses on the Ridge area (Blue in Figure 1). Collectively, these installations form the Greater Clair Area, which includes both developed and undeveloped reservoir zones across multiple horizons.



Clair Development Area and Phases

Clair lies 75 km west of The Shetland Isles on the UK Continental Shelf (UKCS). Scale and complexity drove a multiphase development approach.

The Clair Phase 1 platform came online in 2005 and primarily develops the Horst and Graben areas in addition to other minor blocks.

The Clair Ridge platform came online in 2018 and primarily develops the Ridge area highlighted in blue.

A potential Clair Phase 3 development is under evaluation which would target areas to the south of Phase 1

Figure 1—Clair location and primary development areas.

Reservoir Description

The Devonian age Old Red Sandstone (ORS) reservoir lies 1.5-2.5 km below sea level, sealed by Late Cretaceous mudstones and sourced from the Kimmeridge Clay Formation. The two ORS lithostratigraphic units, the Upper (UCG) and Lower Clair Group (LCG). The LCG, containing the bulk of the oil-in-place, is a sandstone characterized by large variations in facies and permeabilities ranging from <1md to several 100md. The LCG is subdivided into six units, I-VI, based on variations in sedimentary facies and heavy mineral assemblages (Morton, 2003). Development drilling preferentially targets the highest quality reservoirs in Units V and III.

The LCG is defined as a dual-permeability system, with a variable fracture distribution, generally classified as a Type II/Type III fracture reservoir (Nelson, 1981) (Figure 2). Reservoir deliverability and connectivity are primarily controlled by the distribution of existing natural fracture networks. Three main scales of fracture features are observed in core: fault related, fracture corridors unrelated to large offsets and lower density joint-type networks. Open fracture density varies considerably with highest densities in Units V and VI. Open fracture orientation varies between structural compartments (Barr et al., 2007). However, laterals in both Units V and VI have penetrated extensive sections of unfractured or low-fracture-density rock.

Further details can be found in SPE 250250 (Roy, 2022).

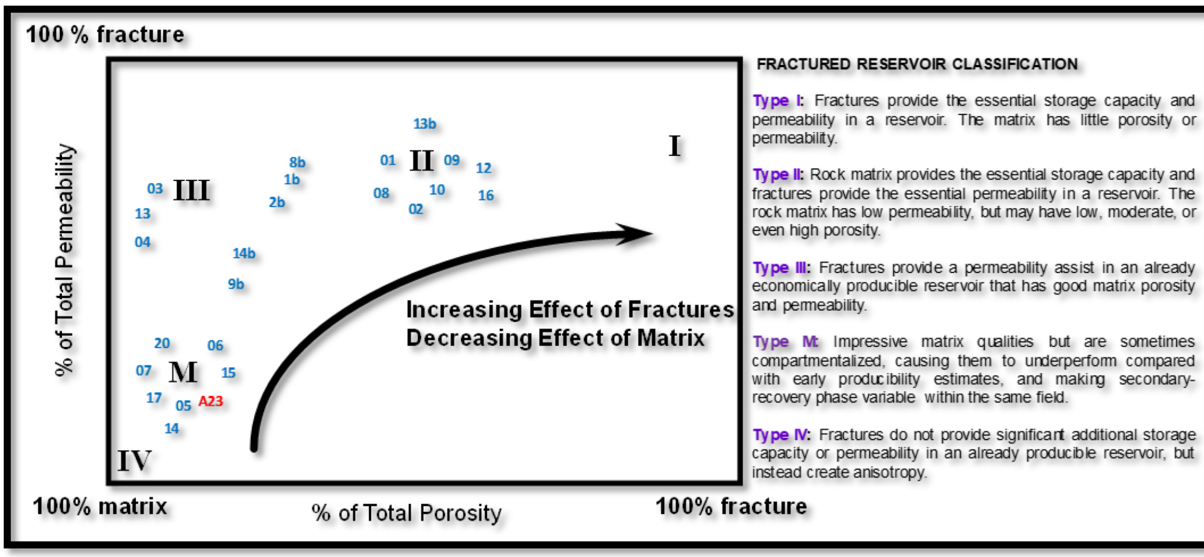


Figure 2—The Clair Field wells categorized as per the fracture classification (in blue wells number).

Field development phases and hydraulic fracturing progress

The scale and complexity of the Clair reservoir drove a phased multiplatform development (Figure 1 and Figure 3). Newer wells incorporate historic lessons ensuring robustness to dynamic uncertainties in this naturally fractured reservoir (NFR). Development wells have encountered varying degrees of natural fractures, the primary control on production outcomes.

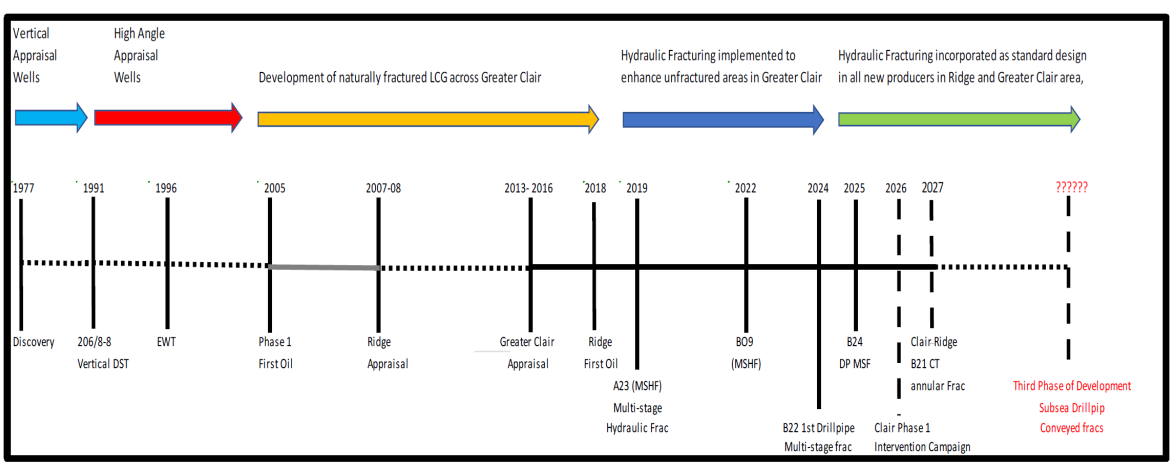


Figure 3—The Greater Clair area development and hydraulic fracturing progression.

Fig 3 provides the current overview of the Clair Field development progressions, updated from the original produced for SPE 205250 after the first multi-stage frac (MSF) in Clair Phase 1 well A23. Two further MSF wells have now been completed on the Clair Ridge Platform, Clair Ridge wells B09 in 2022 and B22 in 2024. The future schedule from 2025 now shows a significant number of planned MSF candidates now hydraulic fracturing is providing a track record of demonstrable uplift in low permeability non-naturally fractured portions of the Clair LCG reservoir. What is also emerging for new drill wells is a standardized completion, based on B22 MSF design utilising drill pipe shifted sleeves. For existing wells, a variety of MSF techniques will continue to be applied on a case-by-case basis, dependent on the liner design, to optimize the executions.

MSF Installed and Planned Installations:

- A23 and B09 – MSF installed via conventional CT shifted sleeves (initial rates 6,500 and 4,500bopd respectively).
- Clair Ridge B22 – MSF via drill pipe shifted sleeves (installed 2024).
- Clair Ridge B24 – MSF via drill pipe shifted sleeves (to be installed 2025).
- Clair Ridge B26 – MSF via drill pipe shifted sleeves (to be installed 2026).
- Clair Ridge Future producers – all Planned as MSF via drill pipe shifted sleeves (B29, B32, B35, B36, B38, B39)
- Clair Ridge B21 - MSF via CT shifted sleeves (2027).
- Clair Phase 1 A07 & A15 – Intervention based campaign in two wells not designed for fracs (planned for 2026).
- Clair Phase 1 – New drilling campaign will include drill pipe shifted sleeve (plan for 2028).
- bp and partners are considering options to unlock future energy potential from the Clair field through a third phase of development. Hydraulic fracturing will be an essential technology with one potential development option being subsea wells with MSF via drill pipe shifted sleeves.

Significant modeling efforts now underpin the potential value of hydraulic fracturing in the Greater Clair and optimization of the designs, further details can be found in IPTC-22293-MS (Dumitrache, 2022), and SPE-212356-MS (Bird 2023).

The primary focus of this current paper is the execution of the completion and fracturing operations in Clair Ridge well B22 which took place in the summer of 2024.

Production Outcomes Through Hydraulic Fracturing

SPE 250250 (Roy, 2022) details how naturally fractured zones and fracture KH dominates wells total deliverability, wells with reduced presence of natural fractures are relatively poor producers. The plan for hydraulic fracturing is therefore to target wells and zones with minimal presence of natural fractures where hydraulic fracturing can increase the production significantly. Hydraulic fracturing provides the ability to establish vertical connectivity in thick reservoir sections between production zones in the horizontal wellbore increasing the total recovery factor in those areas. SPE 215632 details how hydraulic fracturing has delivered short term production increases of up to 6.5-fold and sustained 2 to 3-fold increases in the longer term on both Clair Ridge and Clair Phase 1.

The production outcome for Clair Ridge B22 is the subject of further review in the remainder of this paper, in keeping with the outcomes of the previous MSF installations, the post frac initial production for B22 significantly exceeded the predicted upside P10 prediction however production allocation is ongoing to determine the stimulation uplift and natural fracture contribution.

Completion Design

Original Asset Completion Strategy

The asset's completion strategy for producers revolves around three key objectives:

- Construction - sliding sleeve completion with zonal isolation allowing for multiple hydraulic fractures and different zonal configuration through life of well.
- Stimulation - enabling production from poor quality matrix.
- Life of well – water and/or gas - breakthrough risks.

The naturally fractured nature of the Clair Ridge reservoir, with low anticipated fracture gradients drives the use of open hole completions. It consists of two main units with dual porosity systems and favourable permeability to warrant primary development. Unit III was historically considered to be sand prone (requiring sand control via standalone screens). Unit V was assessed to be more competent and was best completed using a (remotely operated) sliding sleeve completion. Both utilised swell packers for open hole isolation.

These completion designs were initially deemed optimal based on the expected production from the matrix and natural fractures, without the need (and additional cost) of proppant stimulation.

However, targeting natural fractures and higher quality matrix sections of the pay has not resulted in the anticipated production, driving development towards proppant stimulation, to minimise potential downside outcomes.

Reservoir Complexity and its Impact on Completion Design

Standard asset strategy for well placement is to target the best quality rock. Due to the complex nature of the naturally fractured reservoir, the well design of the well requires a number of different assemblies per zone (B22 was installed with 7 zones, 6 being potential stimulation targets). The nature of the wellbore (washed out open holes, high net-to-gross with minimal shale breaks) and presence of several features/ faults/associated damage zones, as well as trajectory azimuth versus the anticipated maximum horizontal stress (longitudinal/oblique fractures), resulted in a sophisticated space out and equipment strategy.

Equipment Needed to Enable Stimulation

Following the change of strategy towards stimulation, for Unit V wells, mechanical stimulation sleeves were introduced and the original remotely operated production sleeves removed. This introduced a start-up intervention. This was not as simple for Unit III wells (where sand control was potentially required). With B22 in Unit III, an engineered solution to enable stimulation (and remote well start-up) in a sand prone environment was developed: remotely operated production sleeves (with and without sand control technology).

To enable online stimulation activities, reliance on swell packers for preventing out-of-zone frac initiation is not possible (as the packers would not have swollen to the required differential/anchoring capacity). So hydraulically set stimulation packers and anchors were introduced. The hydraulic packers provide pressure confinement for initiation and frac generation within the zone of interest. However, the risk of damage to these packers' short elements (during screen-out), made them sub-optimal for life-of-well isolation. The anchors minimize excessive movement, and therefore loads, on the completion string. The swell packers were used to best enhance life-of-well zonal isolation.

Anchoring Capacity in Lower Strength Sands

Due to the lower UCS conditions of the target reservoir, engineering assessments were conducted to determine the most appropriate anchor design for the downhole environment and expected loads. Using materials which could simulate downhole rock strength and then functioning and manipulating the anchors in a test cell (with a maximum oval hole), anchoring capacity was determined. Based on the results, the design with the largest slip contact area were selected.

Equipment Placement

The open hole stimulation strategy comes with its own complexities including unknown initiation points, relying on open hole isolation to confine stimulation pressures, and a large annular space where proppant can accumulate during the stimulation operations.

The placement of these assemblies involved reviewing density, porosity, gamma & UCS for sleeve placement (and likely initiation points), and resistivity, saturation, permeability, image & caliper for packer placement. Losses also informed whether the observed features were open or closed. Deep resistivity tools were run on this well highlighting potential pockets of hydrocarbons away from the wellbore. Placing sleeves close to the likeliest initiation point in the open hole, minimizing the annular path between frac and inflow point. This strategy was used to minimize the risk of annular hole collapse or proppant plug generation.

New Completion Design - Concept Selection, Equipment Selection and Qualification

Offshore fracturing operations require system flexibility in managing operational risks and mitigations to recover from unplanned events, combined with simplicity and efficiency.

Clair does not represent a challenging environment for frac execution per se, stress is mild, pressures relatively low, ability to breakdown formation undisputed, absence of strong stress barriers, low reservoir temperature, ability to place planned amount of proppant quite high with only one early screen out in 13 stages over three wells which did not result in detectable productivity impact. From the frac perspective the only challenge to frac execution is the presence of natural fractures which may result in additional leak off, sometimes identified as pressure dependent leaks off (PDL), which can be addressed pre-emptively with simple frac design modifications, Casero and Stanley 2025.

The two major operational challenges are posed by the harsh weather environment of the West of Shetland Atlantic and the drilling schedule. When combined, these two aspects may result in very short weather window available for frac operations.

The selection of the completion system for Clair considered these environmental conditions and focused on addressing the following key objectives:

1. **Multi-Stage fracturing**. A system/technique able to allow multistage fracturing without the need for multiple trips to perforate or operate the sleeves and setting plugs.
2. **Open Hole System**. Due to the NFR nature of Clair, the system needs to be compatible with uncemented tubulars (i.e., an open hole system in fracturing terminology)
3. **Under-flush Capability**. A system that allows under flushed fracs without the need for additional Coiled Tubing (CT) operations.
4. **Screen Out Management**. A system able to quickly recover from a screen-out without additional CT operations.
5. **Online/Offline fracturing**. A system that could be operated both with CT (offline) and jointed pipes (online) to allow operational flexibility while allowing a quick POP right after drilling rather than waiting for a "rig less" frac as it is common in onshore horizontal multistage wells.

Over the past 25 years the industry has developed multiple systems and techniques that can be used in horizontal multistage wells, Casero, 2021, van Gijtenbeek et al., 2012, Seale, 2006. However, most of them have been targeting shale gas/oil and tight gas North America plays which have a substantially lower permeability, in the nano-Darcy to micro-Darcy range, and whose design relies on over-flushed treatments offset by the large number of fracs (in the hundreds for a standard unconventional well with multiple clusters perforating approach).

Based on the selection criteria set above, the Single Trip MultiStage (STMF) system, (Fig 4)., was selected for Clair development. STMF relies on sleeves, cemented or uncemented, typically operated by CT (offline fracs), Lee et al., 2016, Keong, 2023, or jointed pipe (online fracs). The former are more common in onshore setting and associated primarily with annular pumping, the latter are gaining traction in offshore settings and associated with tubing pumping. The frac string, either CT or DP, remains in hole for the full duration of the frac operations allowing reversing out the excess of proppant in case of screen out, and enabling under-flushed fracs.

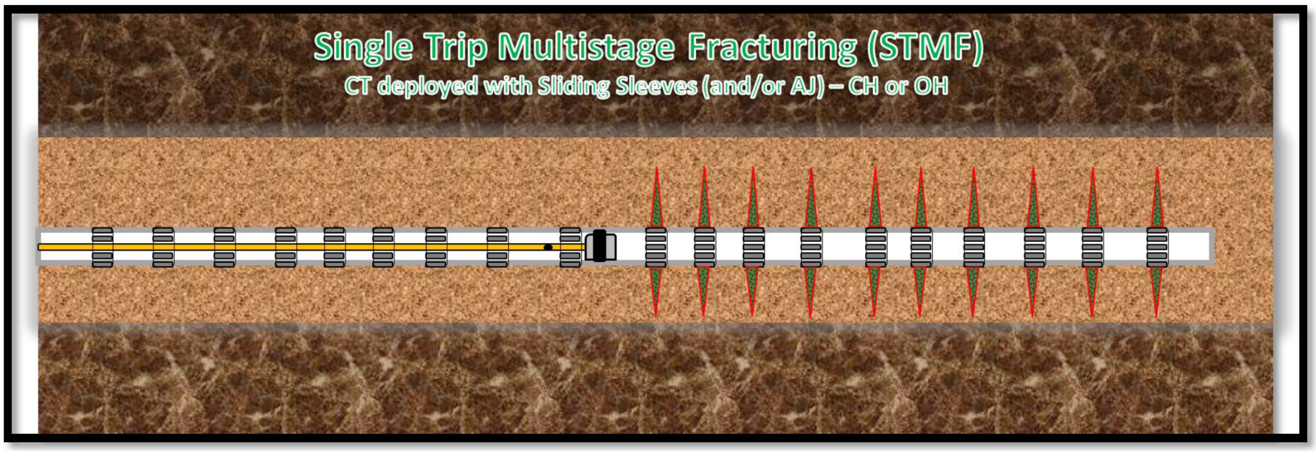


Figure 4—STMF offline annular fracturing simplified schematic (Casero, 2021).

The application of STMF as offline fracturing results in faster stage-to-stage moves but in an offshore setting requires longer mob/de-mob for the CT unit, which makes it not very attractive when the stage count is limited, and CT is not permanently installed on the platform. Conversely, STMF as online fracturing results in slower transition between stages, essentially because of the need for the frac iron rig-up and rig-down at each stage and require strict alignment between drilling and stimulation schedule. This alignment may result in fracturing operations not being executed during the ideal good weather season, this represented a challenge for B22 well and requires adopting other workflow changes to fully and efficiently use the limited and short weather windows available.

The other two main advantages of online fracturing are: first, reduction of deferred production and almost immediate rate acceleration which improves the economics; second, better positioned for Extended Reach Drilling (ERD) wells which might be demanding for smaller CT to reach TD while challenging for annular pumping when larger CT are considered.

Pairing Bespoke Production Sleeves and/or Sand Control Sleeves with Stimulation Sleeves

With B22 targeting the pay zone in Unit III, the historical strategy would have dictated the use of sand screens. However, the original completion design requirements were complex requiring either multi-function sleeves and/or multiple sleeves per zone;

- Ability to stimulate.
- Ability to provide sand control down hole.

- Ability to remotely open and produce the well, post tree installation.

Bespoke production sleeves with remote technology (similar to the sleeves installed on the asset's injectors), with and without a sand screen, were designed for future wells and will be the subject of future SPE papers. The B22 completion was the first online fracturing operation in Clair Ridge therefore the decision was made to minimise operational risk and remove the remote activated production sleeves and sand control sleeves. This allowed the operation to focus on efficient fracturing operations.

The drilling data acquisition plan for B22 allowed the team to acquire the sonic UCS of the formation. Using this data and the regionally agreed sanding risk cut off a reduced requirement for sand control was inferred. Furthermore, offset well analysis of open hole stability of online producers (comparing start-up pressure build-up – PBU – results against later well life conditions) showed the risk of annular collapse/fill and corresponding reduction in production performance was relatively low. Inflow modelling confirmed that with minimal hole fill in the annulus, the impact on production by reducing the number of fracturing/production sleeves per zone was shown to be inconsequential. This decision to remove the requirement for sand control sleeves has been found to be correct for the first 6 months of B22 production, but will be continually assessed over future years.

Execution of Online Stimulation

Platform Rig Up

For online stimulation, the following line-up was constructed, connecting the frac hangers on the intervention deck to a valve manifold on the drill floor. This manifold could then direct flow from the vessel downhole or allowed well contents to be reversed out via the frac hanger to the rig systems. This rig up spanned 25m vertically. The manifold allowed for isolation between the vessel pumping branch and the reversing branch, whilst also allowing for offline flushing of vessel lines to permanent holding tanks on the rig.

The rig floor manifold connected up to hoses which could move dynamically between the different stimulation positions at each stage (opening the sleeve, treatment, closing the sleeve, reversing out, and then pulling out of hole to the next position).

Over-pressure protection existed on the pumping side via pump trips and a gas-operated relief valve on the vessel. On the platform, connected to the wellhead, was an electronic trip via pressure sensors and a burst disk manifold. Downhole pressures were monitored via a "dead-leg" down the annulus (brine column).

On the reverse line, an adjustable choke was used to manage trapped net pressure post-job. Citric acid was injected into the XL gel returns to break any gels and prevent shaker blinding, turbulence was generated via the line length and a strainer and the rig shakers were used to strip out the proppant (sent to skips and weighed). The choke and kill (C&K) manifold and mud gas separator (MGS) were tied in as a contingency routing if required.

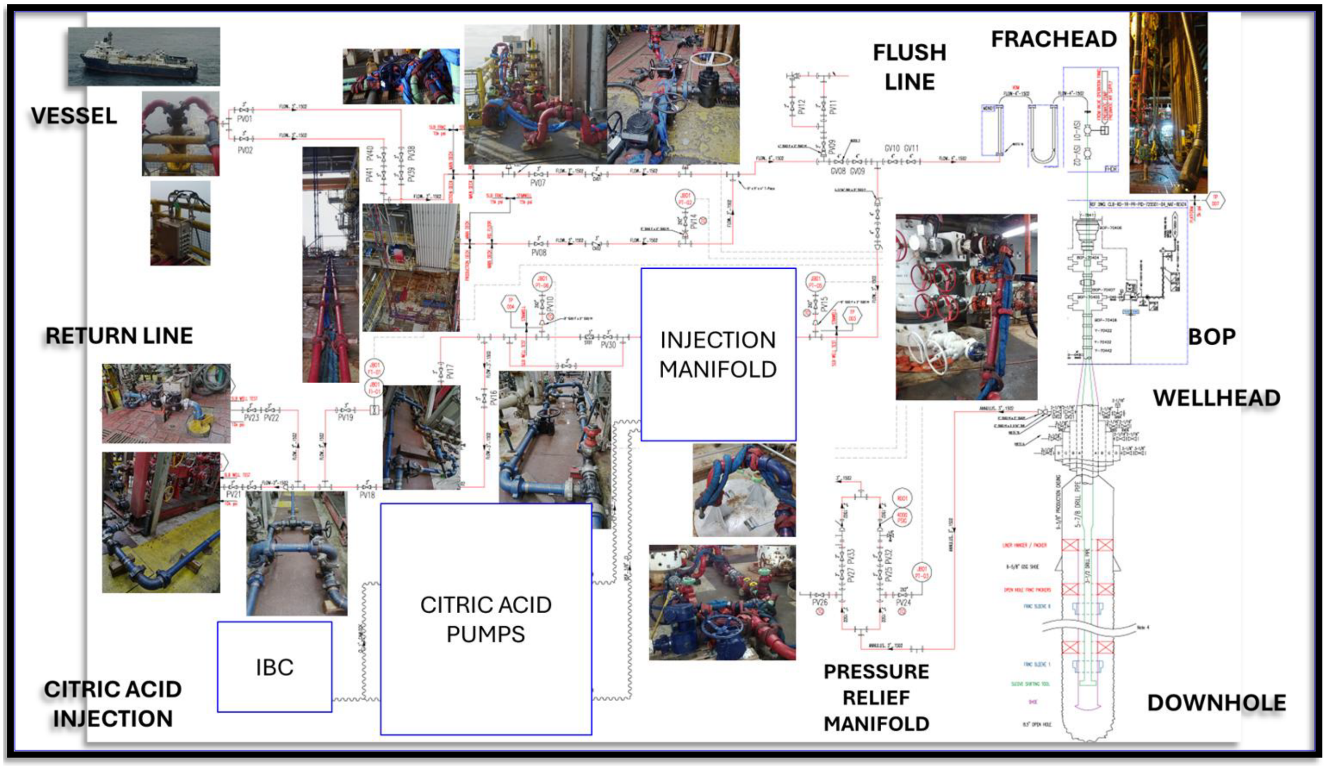


Figure 5—Key surface process blocks (tie-in to shakers for proppant and slurry disposal, C&K/MGS for WC inclusive of hot flush, CRI for vessel flush, citric acid injection spread/strainer, vessel tie in with total vertical height and iron length).

Frachead and Drill Pipe

The frachead crosses over from the rig equipment to the downhole drill pipe. It contained two FOSVs (one actuated manually and the other remotely) and a stripping joint (spaced out across the BOP).

The drill pipe consisted of 5-7/8" drill pipe (with no inner lining) and the inner string (across the lower completion) consisted of 3-1/2" upset drill pipe (with TS6 connections).

Treatment and reverse out pressures were assessed to confirm over-protection requirements and ensure achievable rates to lift the proppant out of the well effectively. The initially selected heavier weight inner string (15.1ppf) resulted in very high treatment pressures. A lighter weight inner string (12.95ppf) reduced treatment pressures. Available flush strings caused torque and drag complexities due to the well length. An assessment of both IDs and ODs was performed to assess the risk of proppant plugging on the backside as well as compatibility with the simulation BHA.

Operational Sequence

The below flowchart outlines the sequence of the rig-based activities (prior to, during and post the online stimulation). Stimulation was completed toe to heel for two main reasons: (i) reduce the risk of having a stuck-open or leaking sleeve above the next zone to be stimulated and (ii) optimize tripping time.

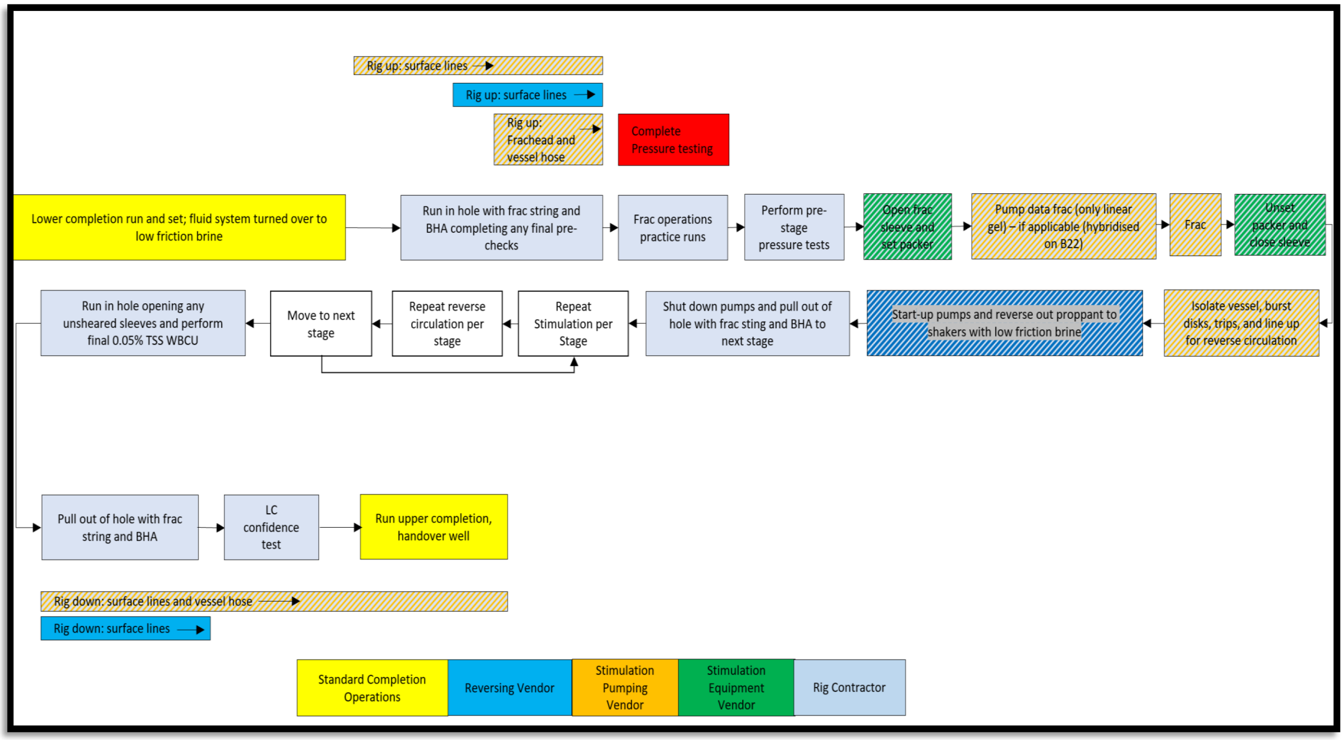


Figure 6—STMF hydraulic fracturing sequence flowchart.

Hydraulic Fracture Treatments Design, Execution and Analysis

Fracture stimulation strategies in the Clair Field are guided by optimization studies aimed at enhancing fracture conductivity (Bird et al., 2023, Dumitrache et al., 2022). To achieve this, larger proppant, specifically 16/20 mesh, is introduced during the final stages of the treatment to improve the conductivity of the proppant pack near the wellbore, where flow efficiency is most critical. The design also incorporates tip screen-out (TSO) techniques and under-displacement to strengthen the connection between the fracture network and the wellbore. To reduce the likelihood of premature screen-out, the initial 33% to 66% of the total proppant volume is composed of finer 20/40 mesh material.

Fluid compatibility testing.

During the proposed online stimulation operations, several different fluids would be in contact with one another. Primarily completion brine (inclusive of boundary layer modifying friction reducer) and XL gel

To ensure compatibility between the completion fluids (primarily brine) as well as the stimulation fluids (broken XL gel) during treatment and reverse out, lab-based fluid testing was required. This was done to analyse the turbidity (solid/emulsion generation), the pH, the oxygen concentration (development of a corrosive environment and general safety risks), and the H2S concentration (SSC and general safety risks). Tests were performed at down-hole temperatures.

No incompatibility issues were identified.

Reverse Out

Overall Strategy

After each treatment, the planned ~5bbl of under-flushed 8ppa slurry was reversed out to the rig (down the annulus and up the frac string). Boundary layer modifying friction reducer (polyacrylamide type fluid) was used to manage circulation pressures due to the length of the well. This was added to the brine used throughout drilling and completion operations.

The time to start reversing out was minimized to ensure no proppant plug formed, causing difficulties reversing out to surface.

Several assumptions were made to constrain the operational steps for reversing out:

- Sleeves were considered closed prior to reverse out (based on track record) but not proven to be sealing based on pressure testing.
- Downhole pressure was attempted to be kept below FCP (fracture closure pressure) to reduce the likelihood of over-displacing the proppant further into the fracture, in case the sleeve was not closed/sealing. This was achieved by starting to reverse at a lower rate.
- Potential fluid ingress from the reservoir during reverse out operations was considered a "ballooning" event, as the injected fluids were supercharged into the formation and so would be the expected fluids entering back into the wellbore (not reservoir fluids).

The operational steps were as per the following flowchart.

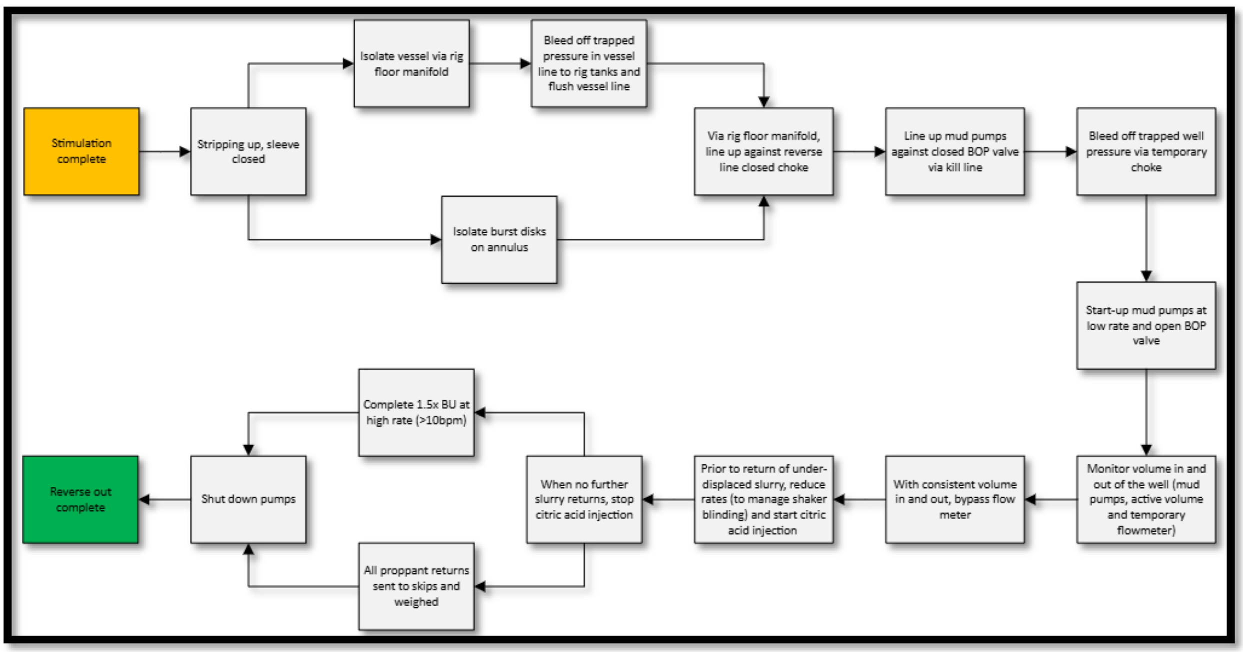


Figure 7—Reversing out operational steps.

Monitoring the Well During Reverse Out Operations

The intent was to use the rig PVT to monitor to monitor wellbore for losses or gains, B22 operations showed that it was challenging due to a reduction in apparent active system volume at the beginning of reverse out associated with lines being filled up and changes in wellbore pressure. At the beginning of each reverse out operation, initial apparent losses were 5-10bbl over ~5-10min. However, when reverse out was conducted with an open sleeve (3rd stage where the sleeve did not close), after the usual 5-10bbl of "apparent" losses, gains were quickly noticeable on rig PVT. However, the well was not shut-in quickly enough, resulting in significant disturbance of the fracture.

B22 data shows it was not possible to accurately monitor the proppant being stripped out over the shakers because of (i) the time it took for the system to respond the change in reverse out rate prior to circulating proppant at surface, (ii) as well as (although to a lower extend) to the corresponding drop in wellbore pressure and (iii) the small amount of proppant which was removed during each reverse out (apart from 3rd stage with influx).

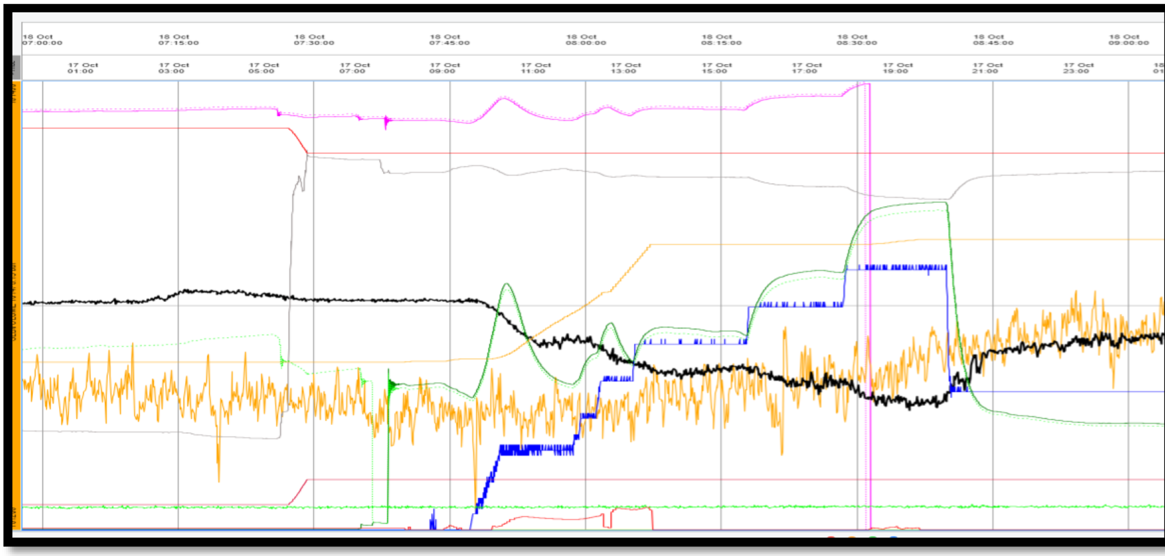


Figure 8—Illustration of how the active system volume (black line) changes during reverse out. Pump rate in blue and annulus surface pressure in green. 1st stage.

Under-displaced Proppant Returns

Across the 5x stages, the quantity of under-flushed proppant returned to surface was very variable: between 0.5 to 5mT (wet proppant), compared to a planned weight of ~0.6mT (for 5bbl of wet proppant). Some of this variance may be due to the inaccurate weighing method (using the platform crane) and different degrees of dehydration.

The injection of 20% citric acid (added at 10gpt), in addition to the line geometry, was efficient to break the gels. Mixing happened over ~15m in 1×3" line at ~3.7bpm. The acid injection strategy was established from lab testing to achieve a quick break out of the gel. The reverse line included one strainer downstream of citric acid injection to help with fluid mixing and shear. However, the strainer was bypassed during several reverse-out and did not impact operations.

During reverse out of the last stage, the under-flushed slurry reached surface significantly earlier than anticipated (~95bbl, compared with a theoretical volume of ~260bbl). No clear reason has been identified for this but may be in part due to stringing out of the slurry with the low viscosity displacement fluid. On two occasions (3rd stage and 5th stage), some XL gel slurry arrived later than anticipated at surface. For the 3rd stage, this is explained by the fact that the reverse out was stopped at some point, which could have allowed heavier XL gel slurry to fall back down the well until the pump was ramped up again. For the 5th stage, where some gel also came out early, gel might have been left behind when reversing at lower rate and was only picked up when the rates were increased (>8bpm).

During the final clean up (performed close to TD, below the bottom stimulation sleeve), there is evidence that the slurry needed a minimum of ~8bpm to be carried to surface throughout all the changes in IDs of the stimulation string.

Annulus and Tubing Pressure Responses When Reversing Out

The presence of XL Gel across the 3.5" pipe led to pressure "spikes" when ramping up the mud pump to reverse out, due to extra friction. Once the gel started to move, the pressure decreased. This was only visible when XL gel was across the 3.5" string, not once it entered the 5-7/8" DP.

The presence of a XL gel "slug" sitting in the 3.5" string led to differences in pressure response between the annulus and drill pipe sides. For example, when pressure was bled on the drill pipe side through the reverse out choke (over 2-3min from ~2000psi to ~140psi), pressure on the annulus side only dropped by

200-900psi over the same time period. The more gel in the 3.5" string, the lower the pressure drop and the slower the response.

There were clear indications of the effect of the friction reducer. When used, pressure was ~1500psi lower when pumping at ~9bpm (~3630psi versus 2140psi).

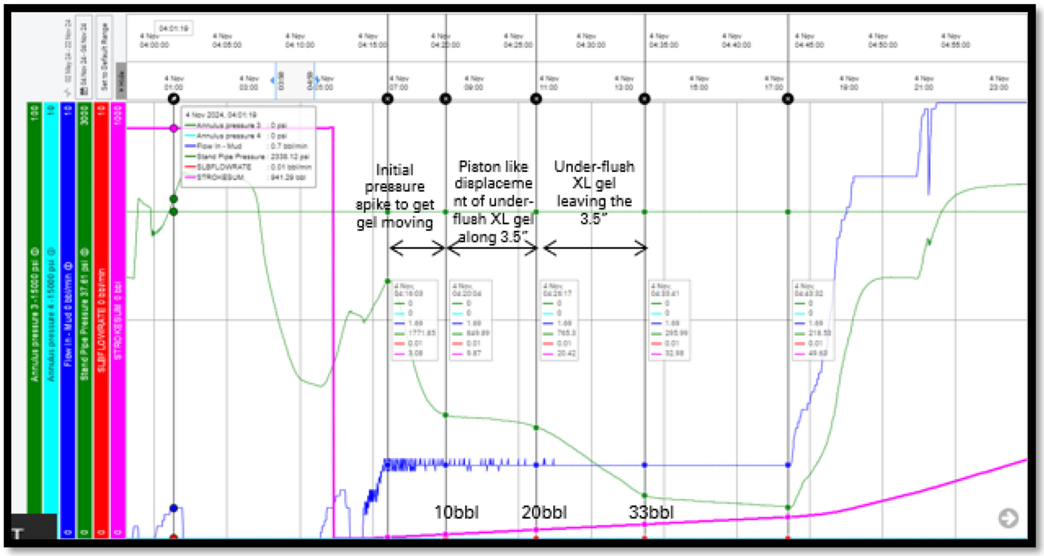


Figure 9—Illustration of pressure drop (green curve) when XL gel existed 3.5" string after 4:26am. 2nd stage.

When Operations Do Not Go As Planned: Sleeve Not Closing Post-Stimulation

On stage #3, the standard operations were followed, but (unknown to the team) the sleeve did not successfully close. Reverse out was started as per plan. The signs of the open sleeve were not detected quickly, allowing a total of 55bbl of slurry to re-enter the wellbore (recorded on rig active system). This volume would have been displaced from the placed fracture.

Once the open sleeve was identified, the stimulation BHA packer was unset and then reset above the open sleeve, isolating it (as per planned contingency). Reverse out operations were continued until the well was confirmed clean.

During reverse out, a total of 20mT of (wet) proppant was recovered at surface. This volume includes the planned ~5bbl of under-flushed slurry and the proppant within the volume that re-entered the wellbore. Considering the average amount of proppant recovered from the 4x other zones, the 55bbl influx had an estimated average dry proppant concentration of ~19ppa.

The volume between the stimulation sleeve and frac mouth is unknown, but if assuming a ~100m (which is the maximum expected distance, placing the frac mouth next the frac packer), the open hole annulus volume would be ~15bbl (assuming an average open ID of 8.75 and completion OD of 5.5"). This means that most of the 55bbl would be coming from inside the fracture itself, which would explain the high average proppant concentration from the de-hydrated proppant. Assuming 8ppa for the 15bbl in the annulus, the reminder volume from the frac would have to be ~26ppa, close to bulk proppant concentration (about 29ppa). The influx could also have contained fluid from the part of the annulus not going to the frac with little to no proppant at all, this would mean that the fluids coming from the frac would have been even more concentrated in proppant (closer to bulk concentration).

While reversing out, dissolved gas up to 1.5% was observed (but no free gas). This may have come from the brine in the annulus not displaced by the treatment, which had more time in contact with reservoir hydrocarbons.

There was no challenge to manage the greater quantity of XL gel and proppant over rig shakers proving the system was effective.

Following this stage, a more detailed description of the monitoring criteria was defined to help the offshore team identify such events during the subsequent stage. They included (i) flowmeter initial rate during bleed down, (ii) flowmeter rate when at constant reverse out rate, (iii) cumulative through the flowmeter, and (iv) active system volume (one expected trend was understood).

Post-Stimulation

Use of boundary layer modifier to reduce PU weights and recover frac string

Following stimulation of all the stages, the BHA was tripped to the bottom sleeve to perform a clean-up and shear open (and then close) sleeves in unstimulated zones. During this clean up, the boundary layer modifying friction reducer was removed from the formulation (as it was no longer expected to be required).

Once the final wellbore cleanup was completed, an attempt to move the string was made. Up weights were significantly more than what had previously been seen and no breakover occurred. The friction reducer was re-introduced, which resulted in the metal-to-metal friction factors dropping back to the same value as had been seen between stages.

A number of root causes were identified: (i) the oily film (from the reservoir drill-in fluid) was removed across the lower completion when pumping the wellbore cleanup pull train. The removal of the slick water exacerbated this. The use of the stimulation BHA for WBCU operations would also have introduced additional drag.

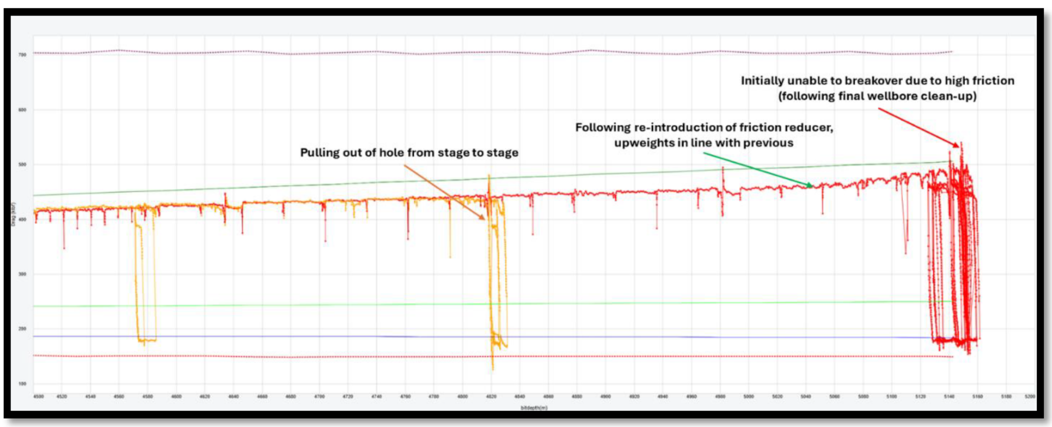


Figure 10—Hookload comparison between pulling out of hole between treatment and pulling out of hole following final wellbore clean-up. Operational Performance

Although this was the instance of online stimulation performed for the asset and organisation (and a learning curve was expected), the performance improvements implemented to simplify the diagnostic stages of the treatment - as well operational experience gained after each stage - resulted in <24 hours per stage.

Zone	Frac Stage	500mz Entry	500mz Exit	Duration
#2	1	17th Oct 11:25	18th Oct 09:35	~22hrs
#3	2	3rd Nov 15:15	7th Nov 10:45	~3days 20hrs
#5	3			
#6	4			
#7	5			

Figure 11—Zone Number and Fracturing Execution Summary

Flowback, Tracers, Downhole Choke and Production Results

A successful well start-up was executed based on gradual bean-up steps (<250psi) over 3-4 days to minimize the risk of proppant production. Well start-up took a phased approach utilising temporary solids management equipment and production test separator to manage both solids and emulsions between oil-based mud and water-based completion/stimulation fluids. Once the water-based fluid rate had dropped below 500bwpd production was then routed to the oil leg for export.

After well start-up, a total of 11.2t of wet proppant was recovered in the platform test separator, estimated as ~9t of dry proppant (~20% of water in wet proppant). For comparison, this would correspond to an open hole annular volume of ~560m, or ~112m/zone (for 5 stimulated zones) of 8ppa slurry. Given the concentration of proppant in 3rd frac stage (zone 5) influx, there's the possibility that 3rd stage would have contributed to a greater proportion of the proppant recovered at surface.

Tracers were only pumped into the stimulated zones (zones 1 & 4 did not have tracers pumped into them). All zones were open for well start up and so understanding absolute contribution from each zone across the entire wellbore was not possible. Tracer analysis is still ongoing, however initial results indicate zones 2, 3, 6 and 7, contributed more evenly than zone 5 (evident from both the gas and oil tracers). Overall, some production was recorded from all stimulated zones during well start-up. However, zone 5 (frac stage 3) stage has very little contribution throughout the current production period. Zone 2, the 1st stimulated stage and closest to the toe, contributes the most with its contribution increasing over time as the well cleaned up.

Lessons Learned and Optimization

Use of Strip Tracers to Understand Contribution

The complexity of the reservoir (dual porosity system) means not all zones are stimulation candidates. On B22, the production from these zones (and therefore the contribution of the stimulated zones) was challenging to estimate. To overcome this, a future well is planned to use strip and inflow tracers to determine contribution of all zones (stimulated or not) to overall well production.

Friction Reducer Optimization

To minimize the risk of excessive drag causing an inability to recover a stimulation string to surface and number of optimizations introduced in future online stimulation wells:

- Replace any wash pills used during/post stimulation with a catch pill, if remaining proppant needs to removed. There is no need for surfactant at this stage.
- Re-order operations to perform any TSS circulations prior to the stimulation.
- Introduce friction reducer (metal-to-metal) during stimulation operations.

Reverse Out Operational Improvements

To improve and simplify the reverse out operations, the following could be considered for future wells:

- Remove strainer from base plan to reduce reverse out pressure and rely on citric acid and reverse line geometry to break gels.
- Citric acid injection point can be brought more upstream in the reverse line to allow more time for mixing between acid and XL in the 3" line.
- Perform a short "dynamic" confidence test prior to bleeding the pressure (via the annulus). Expected pressure response will have to consider the presence of XL gel across the 3.5" pipe which will create a buffer/delay in pressure response between the annulus and the drill pipe.
- Remove the final wellbore clean up at TD as only a small amount of proppant was recovered (compared with the quantity of proppant which flown to surface during well start-up). This

improvement can only be implemented if leaving some proppant inside the lower completion will not impact other activities such as wireline activities.

The Future for Hydraulic Fracturing in the Greater Clair Area

Over the past three years, Clair Field has seen a transformation in its fracturing completions strategy, evolving into a more standardized, simplified system. This evolution has been driven by the need to enhance efficiency in multistage fracturing operations, reduce operational complexity, and mitigate the impact on asset schedules. As the field continues to mature, the future of fracturing in the Greater Clair Area will focus heavily on the strategic transition from offline to online stimulation campaigns and in implementing the online system in Subsea applications in the next phase of development.

Transition to Online Stimulations

One of the most significant advancements is the move from traditional offline stimulations to online stimulations, integrated directly into the rig schedule. Online stimulation offers numerous advantages: they are less disruptive to the wider asset schedule, require fewer personnel onboard (POB), and present a lower SIMOPs burden compared to offline methods. These stimulations are executed during the well's completion phase, typically requiring a small uplift in manpower (4–8 POB) over a 1–3-week window in fair weather conditions.

To support this shift, rig schedules are now being optimized to prioritize stimulations during the more predictable April to September weather window. Although winter windows (Feb–Mar and Oct–Nov) may be considered, this will be evaluated on a case-by-case basis with consideration for weather-related delays and overall program impact.

Flowback Management and Infrastructure Enhancements

A critical enabler for widespread adoption of online stimulations is improved management of the flowback interface with topsides processing and export systems. Online stimulations inherently introduce unflowed wells to the plant, carrying a mix of water and residual oil-based mud that can risk non-conformance with export specifications.

To mitigate this, a dedicated flowback project to reroute test separator output directly into CRI Well, bypassing the main export path has been sanctioned. This will significantly reduce the risk to plant systems and export quality. A temporary solution was implemented for well B22. For Well B24 the main RDF was switched from OBM to WBM which should create less separation issues.

Strategic Scheduling

Future fracturing strategies in Clair also focus on increasing scheduling flexibility, particularly through integration with NWD teams and TAR (Turnaround) activities. Less Operational Efficiency (OE) events and strategic TAR planning help reduce the impact on Integrated Production Capacity (IPC) growth. A final offline stimulation campaign is also being considered to eliminate backlog and serve as a contingency should online stim strategies encounter execution risks.

Enabling the Future

Looking forward, success in the Greater Clair Area will depend on:

- Maximizing online stimulations campaigns to eliminate dependency on offline methods.
- Leveraging schedule integration to deliver value-focused, resilient programs.
- Implementing infrastructure changes or switching the RDF for all producers to WBM to safeguard top-side operations.

- Create learning through repetition an enable execution for future subsea development.

Conclusions and Recommendations

STMF proved to be an effective and flexible system to execute hydraulic fracturing in harsh environment such as the North Atlantic West of Shetland. Its flexibility allowed the successful use the very short windows available late during the year and resulted in placing 5 fracs in 5 days over two weather windows. Such efficiency can be only marginally improved with the online fracturing sequence.

Online fracturing in combination with STMF system was trialed for the first time by bp and its successful deployment was a reflection on the years of planning and testing that preceded the execution.

While the future of fracturing in offshore North Sea depends on many factors, STMF is certainly providing a robust technical base for future applications.

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Nomenclature

bbl	Volume, barrel
bbl/min	Rate, barrel per minute
BOPD	Oil rate, barrel of oil per day
BWPD	Water rate, barrel of water per day
cP	Fluid viscosity, centiPoise
º	Angle, degree
ºF	Temperature, degree Fahrenheit
°C	Temperature, degree Celsius
º API	Oil Density, degree API
ft	Distance, feet
$ft\text{-}min^{-0.5}$	Leak-off coefficient, feet of fluid penetration per square root minute
gal	Volume, US gallon
gpt	Liquid chemical concentration, gallon per thousand gallons of fluid
in	Diameter, inch
in^2	Flow surface, square inch
kg	Mass, kilogram
km	Distance, kilometre
ksi	Stress, thousand pound per square inch
lb/ft^2	Proppant concentration in the fracture, Pound per square foot
lbm	Pound mass
lbm/kgal	Dry polymer concentration, pound mass per thousand gallons of fluid
lbm/ft	Tubular weight in air, pounds mass per linear foot
m	Distance, meter
md	Permeability, Darcy x 10^{-3}
md-ft	Conductivity, proppant pack permeability in millidarcy x frac width in inch
mesh	Diameter, wires per inch
m^3/hr	Rate, cubic meter per hour

mm	Distance, meter x 10^{-3}
mMD/RT	Measured depth below rotary table, meter
MMlbm	Mass, million pounds
MMpsi	Pressure, million pounds per square inch
mTVD	Vertical depth below rig Kelly bushing, meter
mTVDSS	Vertical depth below mean sea level, meter
PPA	Proppant stage concentration, pounds mass of proppant in gallon of clean fluid added
ppt	Dry chemical concentration, pound per thousand gal of fluid
psi	Pressure, pound force per square inch
psi	BHPg Pressure, pound force per square inch recorded at the bottom hole pressure gauge
psi/ft	Pressure gradient, pound force per square in per vertical foot of depth
psi/1000ft	Friction pressure loss gradient, pound force per square inch per linear foot x 10^3
sec-1	Shear rate, reciprocal seconds
sgu	Specific gravity, gram per cubic centimetre
spf	Perforation density, shots per linear foot
t	Mass, kilogram x 10^3
ACA	After closure analysis
BH	Bottomhole
BHA	Bottomhole assembly
BHP	Bottomhole pressure, psi
BHPg	Bottomhole pressure at gauge depth, psi
BHST	Bottomhole static temperature, ºF
CCL	Casing-collar locator
CT	Coiled tubing
DP	Dynamic position
EWT	Extended well test
frac	Fracture
F_{cd}	Dimensionless fracture conductivity,
FOI	Fold of increase
FS	Fracture stage
GLV	Gas Lift Valve
geo	Geometric mean
H_f	Fracture height, m
HTHP	High temperature high pressure
ID	Internal diameter, in.
ISIP	Instantaneous shut-in pressure, psi
ISO	International Organization for Standardization
K_H	Horizontal permeability, md
K_V	Vertical permeability, md
KH	formation permeability x sand height, md-ft
LCG	Lower Clair Group
LHS	Left hand side
LT	Low temperature
LW	Light weight
LWP	Light Weight Proppant
NFR	Naturally Fractured Reservoirs
NWB	Near wellbore

NPT	Non-productive time
OIM	Offshore installation manager
OWC	Oil-Water Contact
PAD	Fracture initiation and propagation stage preceding proppant slurry stages
PBR	Polished Bore Receptacle
PBU	Pressure build up
P_c	Closure Pressure, psi
PCA	Post Closure Analysis
PLT	Production logging tool
P_{Net}	Net Pressure, psi
POB	Person on board
POOH	Pull out of hole
RCP	Resin coated proppant
RHS	Right hand side
RIH	Run in Hole
SDT	Stepdown test
S_H	Maximum principal horizontal stress, psi
Sh	Minimum principal horizontal stress, psi
SIT	System Integration Test
SOR	Statement of requirement
STMF	Single Trip Multistage Frac
STOIIP	Stock-tank oil initially in place
SRT	Step rate test
UCS	Unconfined compressive stress, psi
UKCS	United Kingdom continental shelf
WEG	Wireline entry guide
WH	Well head
WHP	Wellhead pressure, psi
X_f	Fracture Half-Length, m

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